

Group Theory

From groups to quotient groups

Discrete Structures · Tutorial

Today's path



Foundations

groups, basic lemmas



Subgroups

tests and examples



Cosets

Lagrange's theorem



Normal subgroups

quotient groups



Homomorphisms

kernel, First Isomorphism



Cyclic groups

their subgroups

Binary operation & closure

A binary operation on G takes $a, b \in G$ and returns $a \cdot b$. Closure: $a \cdot b$ must land in G again.

Closed: $(\mathbb{Z}, +)$ and $(\mathbb{Z}, \times)$

Not closed: subtraction on positive integers ($2 - 5 = -3$)

Not closed: division on nonzero integers ($1 \div 2$)

Group

A set G with an operation $\cdot$ is a group if these three hold:

1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. Identity: some e with $a \cdot e = e \cdot a = a$
3. Inverses: every a has some b with $ab = ba = e$

Closure is already built into 'binary operation'.

Abelian group

A group is Abelian if $ab = ba$ for all $a, b \in G$.

Abelian: $\mathbb{Z}$, $\mathbb{Q}^*$, $\mathbb{R}^*$ and $\mathbb{Z}_n$ under their usual operations

Non-abelian: $GL(2, \mathbb{R})$, invertible 2×2 matrices

Non-abelian: D_4 , the symmetries of a square ($HD \neq DH$)

Famous examples: group or not?

Set and operation	Group?	Reason
$(\mathbb{Z}, +)$	Yes	identity 0, inverse $-a$
$(\mathbb{Z}, \times)$	No	2 has no inverse in $\mathbb{Z}$
$(\mathbb{Q}^*, \times)$	Yes	identity 1, inverse $1/a$
$(\mathbb{Z}, -)$	No	$(5-3)-2 \neq 5-(3-2)$
$(\mathbb{Z}_n, + \bmod n)$	Yes	inverse of j is $n - j$
(odd integers, +)	No	not closed, no 0

Cayley table of $U(10)$

$\times \bmod 10$	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

$$U(10) = \{1, 3, 7, 9\}$$

$$7 \times 9 = 63 \equiv 3$$

Identity is 1

3 and 7 are inverses

Each row and column has every element exactly once

Cancellation law

If $ba = ca$ then $b = c$. Likewise $ab = ac \Rightarrow b = c$.

Proof

$$(ba)a^{-1} = (ca)a^{-1}$$

$$b(aa^{-1}) = c(aa^{-1}) \quad \text{by associativity}$$

$$be = ce, \text{ so } b = c$$

The left-hand version: multiply on the left by a^{-1} .

Identity is unique

A group has exactly one identity element.

Proof

Suppose e and f are both identities.

$e \cdot f = f$ because e is an identity

$e \cdot f = e$ because f is an identity

Hence $e = f$.

Inverses are unique

Every $a \in G$ has exactly one inverse.

Proof

Suppose $ax = ay = e$, and let b satisfy $ba = e$.

$$x = ex = (ba)x = b(ax) = b(ay) = (ba)y = ey = y$$

So the inverse is unique.

Inverse of an inverse

$$(a^{-1})^{-1} = a \text{ for every } a \in G.$$

Proof

$$a^{-1} \cdot a = e = a \cdot a^{-1}$$

So a is an inverse of a^{-1} .

Inverses are unique, hence $(a^{-1})^{-1} = a$.

Inverse of a product

$$(ab)^{-1} = b^{-1}a^{-1} \text{ for all } a, b \in G.$$

Proof

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = a \cdot e \cdot a^{-1} = e$$

Uniqueness of inverses finishes it.

Socks and shoes: undo in the reverse order.

Two classic exercises

Q1. Show $(ab)^2 = a^2b^2$ if and only if $ab = ba$.

$a(ba)b = a(ab)b \Leftrightarrow ba = ab$ by cancellation on both sides.

Q2. If $x^2 = e$ for every x , show G is Abelian.

$ab = (ab)^{-1} = b^{-1}a^{-1} = ba$, since every element is its own inverse.

Subgroup

A non-empty subset H of G is a subgroup ($H \leq G$) if H is itself a group under G 's operation.

Always present: $\{e\}$ and G itself

Example: $5\mathbb{Z} \leq \mathbb{Z}$ (multiples of 5)

Non-example: odd integers in $\mathbb{Z}$ ($1 + 1 = 2$ is not odd)

Subgroup test

$H \neq \emptyset$ is a subgroup of G if and only if (1) $a, b \in H \Rightarrow ab \in H$ and (2) $a \in H \Rightarrow a^{-1} \in H$.

Why this is enough

For $a \in H$: $a^{-1} \in H$, so $aa^{-1} = e \in H$

Associativity is inherited from G

One-step version: $a, b \in H \Rightarrow ab^{-1} \in H$

Finite closure criterion

If H is finite, non-empty and closed under the operation, then $H \leq G$.

Proof

For $a \in H$: $a, a^2, a^3, \dots$ all lie in H

H is finite, so $a^r = a^s$ for some $r > s > 0$

Cancel: $a^{(r-s)} = e$, so $e \in H$

Then $a \cdot a^{(r-s-1)} = e$, so $a^{-1} = a^{(r-s-1)} \in H$

Is it a subgroup?

$\{0, 4, 8, 12\}$ in $\mathbb{Z}_{16}$ under $+$: Yes. Finite and closed.

$\{0, 4, 8, 12\}$ under $+$ mod 15: No. $8 + 8 = 16 \equiv 1$ is missing.

$\{a + bi : a^2 + b^2 = 1\}$ in $\mathbb{C}^*$ under $\times$: Yes. $|zw| = |z| |w|$ and $|1/z| = 1$.

Upper triangular invertible 2×2 matrices in $GL(2, \mathbb{R})$: Yes.

Intersection of subgroups

If $H, K \leq G$, then $H \cap K \leq G$.

Proof

$e \in H$ and $e \in K$, so $H \cap K \neq \emptyset$

Take $a, b \in H \cap K$. Then $ab^{-1} \in H$ and $ab^{-1} \in K$

So $ab^{-1} \in H \cap K$ and the subgroup test applies

A union usually fails: $2\mathbb{Z} \cup 3\mathbb{Z}$ contains 2 and 3 but not 5.

Centralizer of an element

For fixed $a \in G$, $N(a) = \{x \in G : xa = ax\}$ is a subgroup of G .

Proof

$e \in N(a)$.

$x, y \in N(a)$: $(xy)a = x(ya) = x(ay) = (xa)y = (ax)y = a(xy)$

$x \in N(a)$: multiply $xa = ax$ by x^{-1} on both sides to get $ax^{-1} = x^{-1}a$

A subgroup absorbs its elements

If $H \leq G$ and $h \in H$, then $hH = H$.

Proof

$hH \subseteq H$: each hx with $x \in H$ is a product of elements of H

$H \subseteq hH$: for $x \in H$ write $x = h(h^{-1}x)$

and $h^{-1}x \in H$, so $x \in hH$

Congruence modulo H

For $a, b \in G$: $a \equiv b \pmod{H}$ means $ab^{-1} \in H$.

In $(\mathbb{Z}, +)$ with $H = n\mathbb{Z}$, ab^{-1} becomes $a - b$

So $a \equiv b \pmod{H}$ is ordinary congruence mod n

Same idea, now for any group and any subgroup.

Congruence is an equivalence relation (1/2)

$a \equiv b \pmod{H}$ is reflexive, symmetric and transitive.

Reflexive and symmetric

Reflexive: $aa^{-1} = e \in H$, so $a \equiv a$

Symmetric: if $ab^{-1} \in H$, then its inverse $(ab^{-1})^{-1} = ba^{-1} \in H$

So $a \equiv b$ implies $b \equiv a$

Congruence is an equivalence relation (2/2)

Transitive

Suppose $a \equiv b$ and $b \equiv c$.

Then $ab^{-1} \in H$ and $bc^{-1} \in H$

Their product $(ab^{-1})(bc^{-1}) = ac^{-1}$ is in H

So $a \equiv c$.

Hence G splits into equivalence classes.

Right and left cosets

For $H \leq G$ and $a \in G$: $Ha = \{ha : h \in H\}$ and $aH = \{ah : h \in H\}$.

Example: $G = \mathbb{Z}$, $H = 3\mathbb{Z}$

$H + 1 = \{ \dots, -2, 1, 4, 7, \dots \}$

a itself is always in Ha , because $e \in H$

A coset is an equivalence class

The class of a is exactly the coset Ha : $\{x : x \equiv a\} = Ha$.

Proof

$$x \equiv a \Leftrightarrow xa^{-1} \in H$$

$$\Leftrightarrow xa^{-1} = h \text{ for some } h \in H$$

$$\Leftrightarrow x = ha \text{ for some } h \in H \Leftrightarrow x \in Ha$$

Cosets are equal or disjoint

Two right cosets Ha and Hb are either identical or have no element in common.

Proof

Suppose $c = h_1a = h_2b$ is a common element.

Then $a = h_1^{-1}h_2b$

So $ha = (hh_1^{-1}h_2)b \in Hb$ for every $h \in H$, giving $Ha \subseteq Hb$

The same argument gives $Hb \subseteq Ha$.

All cosets have the same size

There is a one-to-one correspondence between any two right cosets of H .

Proof: define $\varphi(ha) = hb$

Well-defined: $ha = h'a \Rightarrow h = h'$

Injective: $hb = h'b \Rightarrow h = h'$

Surjective: hb is the image of ha

So every coset has exactly $|H|$ elements.

Lagrange's theorem

If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$.

$H \cdot |H|$ elements

$Ha \cdot |H|$ elements

$Hb \cdot |H|$ elements

Cosets cut G into equal, non-overlapping pieces: $|G| = 3 \times |H|$

Proof of Lagrange's theorem

1. Every $g \in G$ lies in a coset: $g \in Hg$.
2. Distinct right cosets are disjoint, so they partition G .
3. Each coset has exactly $|H|$ elements.
4. If there are k distinct cosets, then $|G| = k \cdot |H|$.

Therefore $|H|$ divides $|G|$.

Index of a subgroup

The index $[G : H]$ is the number of distinct right cosets of H in G .

$$[G : H] = |G| / |H|$$

Example: $G = \mathbb{Z}_{12}$, $H = \{0, 4, 8\}$

$$|G| / |H| = 12 / 3 = 4 \text{ cosets}$$

$H, H+1, H+2, H+3$

Groups of prime order

If $|G| = p$ is prime, then G is cyclic.

Proof

Pick $a \neq e$. The subgroup $\langle a \rangle$ has more than one element.

By Lagrange, $|\langle a \rangle|$ divides p , so $|\langle a \rangle| = p$

Hence $\langle a \rangle = G$.